C Programming
Platform Coding Practice Question Bank

Module 1 — Input/output, variables and basic arithmetic

No.	Problem (click to open)	Platform	Difficulty
1	Read from STDIN, Write to STDOUT	HackerEarth	Very Easy
2	Solve Me First	HackerRank	Very Easy
3	2235 — Add Two Integers	LeetCode	Very Easy
4	2651 — Calculate Delayed Arrival Time	LeetCode	Easy
5	2739 — Total Distance Traveled	LeetCode	Easy

Module 2 — Conditional statements and operators

No.	Problem (click to open)	Platform	Difficulty
1	Conditional Statements in C	HackerRank	Very Easy
2	2413 — Smallest Even Multiple	LeetCode	Very Easy
3	Grading Students	HackerRank	Easy
4	Roy and Profile Picture	HackerEarth	Easy
5	FizzBuzz	HackerRank	Easy

Module 3 — Loops and repeated computation

No.	Problem (click to open)	Platform	Difficulty
1	For Loop in C	HackerRank	Very Easy
2	1342 — Number of Steps to Reduce a Number to Zero	LeetCode	Easy
3	2169 — Count Operations to Obtain Zero	LeetCode	Easy
4	1523 — Count Odd Numbers in an Interval Range	LeetCode	Easy
5	2652 — Sum Multiples	LeetCode	Easy
6	Number Line Jumps	HackerRank	Easy–Moderate

Module 4 — Digit extraction and number manipulation

No.	Problem (click to open)	Platform	Difficulty

1	Digit Sum	HackerEarth	Easy
2	Find Digits	HackerRank	Easy
3	1281 — Subtract the Product and Sum of Digits	LeetCode	Easy
4	2520 — Count the Digits That Divide a Number	LeetCode	Easy
5	9 — Palindrome Number	LeetCode	Easy–Moderate
6	7 — Reverse Integer	LeetCode	Moderate

Module 5 — Factors, divisibility and mathematical loops

No.	Problem (click to open)	Platform	Difficulty
1	231 — Power of Two	LeetCode	Easy
2	326 — Power of Three	LeetCode	Easy
3	Factorial!	HackerEarth	Easy
4	Count Divisors	HackerEarth	Easy–Moderate
5	Prime Numbers	HackerEarth	Easy–Moderate
6	507 — Perfect Number	LeetCode	Easy–Moderate

Module 6 — Nested loops: online judge problems

No.	Problem (click to open)	Platform	Difficulty
1	Staircase	HackerRank	Easy
2	Printing Pattern Using Loops	HackerRank	Moderate

Module 7 — Nested-loop and pattern-printing questions

Question 1 — Print a Solid Square

Difficulty: Very Easy

Problem statement: Given a positive integer n, print a square containing n rows and n columns of asterisks (*). Do not print spaces between asterisks.

Input format: One integer n (1 ≤ n ≤ 20).

Output format: Print n lines, each containing exactly n asterisks.

Sample Input 4	**Sample Output** **** **** **** ****

Hint: Use one loop for rows and one loop for columns.

Question 2 — Print a Right-Angled Triangle

Difficulty: Very Easy

Problem statement: Given n, print a left-aligned triangle. Row i must contain exactly i asterisks, for i from 1 to n.

Input format: One integer n (1 ≤ n ≤ 20).

Output format: Print n lines. Line i contains i asterisks.

Sample Input 4	**Sample Output** * ** *** ****

Hint: The inner loop limit depends on the current row.

Question 3 — Print an Inverted Triangle

Difficulty: Easy

Problem statement: Given n, print a left-aligned inverted triangle. The first row contains n asterisks; each following row contains one fewer.

Input format: One integer n (1 ≤ n ≤ 20).

Output format: Print n lines, with the number of asterisks decreasing from n to 1.

Sample Input 4	**Sample Output** **** *** ** *

Hint: For row i, print n - i + 1 asterisks.

Question 4 — Print a Number Triangle

Difficulty: Easy

Problem statement: Given n, print n rows. On row i, print integers from 1 through i, separated by one space.

Input format: One integer n (1 ≤ n ≤ 20).

Output format: Print n lines with numbers separated by one space and no trailing spaces.

Sample Input	Sample Output
4	1 1 2 1 2 3 1 2 3 4

Hint: The inner loop prints from 1 through the current row number.

Question 5 — Print a Repeated-Number Triangle

Difficulty: Easy

Problem statement: Given n, print n rows. On row i, print the number i exactly i times, separated by one space.

Input format: One integer n (1 ≤ n ≤ 20).

Output format: Print n lines; row i contains i copies of i.

Sample Input	Sample Output
4	1 2 2 3 3 3 4 4 4 4

Hint: Print the row counter during each inner-loop iteration.

Question 6 — Print a Consecutive-Number Triangle

Difficulty: Easy

Problem statement: Print consecutive positive integers across triangular rows. Row 1 has one number, row 2 has two, and so on. Counting starts at 1 and continues across rows.

Input format: One integer n (1 ≤ n ≤ 20).

Output format: Print n lines, with values separated by one space.

Sample Input	Sample Output
4	1 2 3 4 5 6 7 8 9 10

Hint: Maintain one counter that increases after each printed number.

Question 7 — Print a Right-Aligned Star Triangle

Difficulty: Easy

Problem statement: Given n, print a right-aligned triangle. Row 1 has one asterisk and row n has n asterisks. Use leading spaces to align the right edges.

Input format: One integer n (1 ≤ n ≤ 20).

Output format: Print n lines. Leading spaces are required; do not add trailing spaces.

Sample Input	Sample Output
4	 * ** *** ****

Hint: Row i has n - i leading spaces and i asterisks.

Question 8 — Print a Centered Pyramid

Difficulty: Easy–Moderate

Problem statement: Given n, print a centered pyramid with n rows. Row i contains 2i - 1 asterisks, preceded by enough spaces to center it.

Input format: One integer n (1 ≤ n ≤ 20).

Output format: Print n lines of leading spaces followed by an odd number of asterisks.

Sample Input
4
	Sample Output
	`   *`
	`  ***`
	`*****`
	`*******`

Hint: For row i, print n - i spaces and then 2*i - 1 asterisks.

Question 9 — Print an Inverted Centered Pyramid

Difficulty: Easy–Moderate

Problem statement: Given n, print an inverted centered pyramid. The first row contains 2n - 1 asterisks; each next row has two fewer and one more leading space.

Input format: One integer n (1 ≤ n ≤ 20).

Output format: Print n lines, with the pyramid narrowing toward the bottom.

Sample Input
4
	Sample Output
	`*******`
	`  *****`
	`   ***`
	`    *`

Hint: For row i, print i - 1 spaces and 2*(n - i) + 1 asterisks.

Question 10 — Print a Hollow Square

Difficulty: Moderate

Problem statement: Given n, print an n-by-n hollow square. Print an asterisk on the first row, last row, first column, or last column. Print a space inside.

Input format: One integer n (1 ≤ n ≤ 20).

Output format: Print n lines of width n; interior positions are spaces.

Sample Input
5
	Sample Output
	`*****`
	`*   *`
	`*   *`
	`*   *`
	`*****`

Hint: Print '*' when the row or column is on a boundary; otherwise print a space.

Question 11 — Print a Hollow Right-Angled Triangle

Difficulty: Moderate

Problem statement: Given n, print the outline of a left-aligned right-angled triangle. Print asterisks in the first column, on the diagonal boundary, and across the final row. Print spaces inside.

Input format: One integer n (1 ≤ n ≤ 20).

Output format: Print n lines. Preserve interior spaces; the final row contains n asterisks.

Sample Input	Sample Output
5	* ** * * * * *****

Hint: A position is on the outline if it is in the first column, on the diagonal, or in the final row.

Question 12 — Print a Multiplication Table Grid

Difficulty: Moderate

Problem statement: Given n, print an n-by-n multiplication grid. At row i and column j, print i multiplied by j. Separate values with one space.

Input format: One integer n (1 ≤ n ≤ 12).

Output format: Print n lines, each containing n products separated by one space.

Sample Input	Sample Output
3	1 2 3 2 4 6 3 6 9

Hint: The value at each position is the row number multiplied by the column number.

Question 13 — Print a Checkerboard

Difficulty: Moderate

Problem statement: Given n, print an n-by-n checkerboard using X and O. The top-left character must be X; adjacent horizontal and vertical cells alternate.

Input format: One integer n (1 ≤ n ≤ 20).

Output format: Print n lines of n characters, without spaces between characters.

Sample Input	Sample Output
4	X0X0 0X0X X0X0 0X0X

Hint: Use whether the sum of the row and column indices is even or odd.

Question 14 — Print Floyd's Triangle

Difficulty: Easy–Moderate

Problem statement: Given n, print Floyd's triangle with n rows. Fill it with consecutive positive integers starting at 1, placing one number on row 1, two on row 2, and so on.

Input format: One integer n (1 ≤ n ≤ 20).

Output format: Print n lines with numbers separated by one space.

Sample Input	Sample Output
4	1 2 3 4 5 6 7 8 9 10

Hint: Use a counter that continues increasing across rows.

Question 15 — Print a Butterfly Pattern

Difficulty: Moderate

Problem statement: Given an odd integer n, print a butterfly made of asterisks. The top half widens row by row and the lower half mirrors it. Use spaces between the left and right wings.

Input format: One odd integer n (3 ≤ n ≤ 19).

Output format: Print n lines. Each line contains left and right wings with a central gap.

Sample Input 5	**Sample Output** * * ** ** *** *** ** ** * *

Hint: Build the upper half with increasing wing sizes, then mirror it.

Question 16 — Print a Diamond Pattern

Difficulty: Moderate

Problem statement: Given n, print a centered diamond. The upper half has n rows including the widest row; then print n - 1 rows in decreasing width.

Input format: One integer n (1 ≤ n ≤ 10), representing the number of rows in the upper half.

Output format: Print 2n - 1 centered lines.

Sample Input 3	**Sample Output** * *** ***** *** *

Hint: Print an increasing pyramid, then a decreasing half without repeating the widest row.

Final submission checklist

- Use C and write a complete program for every custom pattern question.
- Use nested loops to decide what each row and column prints.
- Match spaces and line breaks exactly as shown in the sample output.
- Test boundary inputs such as n = 1 and the largest permitted n.
- For online judge problems, read the official problem statement and input constraints before coding.